



* Practical - 19 *

* activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/root"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <!-- Title bar -->
    <LinearLayout
        android:orientation="horizontal"
        android:padding="8dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <EditText
            android:id="@+id/etURL"
            android:hint="Enter URL e.g. https://example.com"
            android:inputType="textUri"
            android:singleLine="true"
            android:layout_weight="1"
            android:layout_width="0dp"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/btnGo"
            android:text="Go"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
    </LinearLayout>

    <!-- Controls -->
    <LinearLayout
        android:orientation="horizontal"
        android:paddingStart="8dp"
        android:paddingEnd="8dp"
        android:paddingBottom="8dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <Button
            android:id="@+id/btnBack"
            android:text="◀ Back"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/btnForward"
            android:text="Forward ▶"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/btnReload"
            android:text="Reload"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/btnClearHistory"
            android:text="Clear History"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"/>
    </LinearLayout>

    <!-- Tiny progress bar -->
    <ProgressBar
        android:id="@+id/progress"
        style="@android:attr/progressBarStyleHorizontal"
        android:max="100"
        android:visibility="gone"
        android:layout_width="match_parent"
        android:layout_height="3dp"/>

    <!-- Web content -->
    <WebView
        android:id="@+id/webView"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"/>
    </LinearLayout>

```




* MainActivity.java

```
package com.example.webbrowser;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    private WebView webView;
```

```
    private EditText etUrl;
```

```
    private Button btnGo, btnBack, btnForward,  
                btnReload, btnClearHistory;
```

```
    private ProgressBar progress;
```

```
    @SuppressWarnings("SetJavaScriptEnabled")
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        webView = findViewById(R.id.webView);
```

```
        etUrl = findViewById(R.id.etUrl);
```

```
        btnGo = findViewById(R.id.btnGo);
```

```
        btnBack = findViewById(R.id.btnBack);
```

```
        btnForward = findViewById(R.id.btnForward);
```

```
        btnReload = findViewById(R.id.btnReload);
```

```
        btnClearHistory = findViewById(R.id.btnClearHistory);
```

```
        progress = findViewById(R.id.progress);
```

```
        WebSettings ws = webView.getSettings();
```

```
        ws.setJavaScriptEnabled(true);
```

```
        ws.setDomStorageEnabled(true);
```

```
        ws.setLoadWithOverviewMode(true);
```



```
ws.setUseWideViewPort(true);  
ws.setBuildInZoomControls(true);  
ws.setDisplayZoomControls(false);
```

```
webView.setWebViewClient(new WebViewClient() {  
    @Override public void onPageStarted (WebView  
        view, String url, Bitmap favicon) {  
        progress.setVisibility(View.VISIBLE);  
        etUrl.setText(url);  
        updateNavButtons();  
    }  
});
```

```
    @Override public void onPageFinished (WebView  
        view, String url) {  
        progress.setVisibility(View.GONE);  
        updateNavButtons();  
    }  
});
```

```
});
```

```
webView.setWebChromeClient(new WebChromeClient() {  
    @Override public void onProgressChanged (WebView  
        view, int newProgress) {  
        progress.setProgress(newProgress);  
        if (newProgress >= 100) progress.setVisibility  
            (View.GONE);  
    }  
});
```

```
});
```



```
btnGo.setOnClickListener (v -> loadFromBar());
etUrl.setOnEditorActionListener (Cv, actionId,
                                event) -> {
```

```
    if (actionId == EditorInfo.IME_ACTION_GO ||
        actionId == EditorInfo.IME_ACTION_DONE) {
        loadFromBar();
        return true;
    }
```

```
    return false;
});
```

```
btnBack.setOnClickListener (v -> {
```

```
    if (webView.canGoBack()) webView.goBack();
    updateNavButtons();
```

```
});
```

```
btnForward.setOnClickListener (v -> {
```

```
    if (webView.canGoForward()) webView.goForward();
    updateNavButtons();
```

```
});
```

```
btnReload.setOnClickListener (v -> webView.reload());
```

```
btnClearHistory.setOnClickListener (v -> {
```

```
    webView.clearHistory();
```

```
    webView.clearCache(true);
```

```
    CookieManager.getInstance().removeAllCookies(null);
```

```
    CookieManager.getInstance().flush();
```

```
    updateNavButtons();
```

```
    Toast.makeText(this, "History & cache cleared",
```

```
        Toast.LENGTH_SHORT).show();
```

```
});
```



```
String home = "https://www.example.com";
etUrl.setText(home);
webView.loadUrl(home);
}

private void loadFromBar() {
    String url = etUrl.getText().toString().trim();
    if (TextUtils.isEmpty(url)) {
        Toast.makeText(this, "Enter a URL",
            Toast.LENGTH_SHORT).show();
        return;
    }

    if (!url.startsWith("http://") && !url.startsWith(
        "https://")) {
        url = "https://" + url;
    }
    webView.loadUrl(url);
}

private void updateNavButtons() {
    btnBack.setEnabled(webView.canGoBack());
    btnForward.setEnabled(webView.canGoForward());
}

⑨ Override
public void onBackPressed() {
    if (webView.canGoBack()) {
        webView.goBack();
        updateNavButtons();
    } else {
```




```
        super.onBackPressed();  
    }  
}
```

⑨ Override

```
protected void onDestroy() {  
    if (webView != null) {  
        webView.loadUrl("about:blank");  
        webView.clearHistory();  
        ((android.view.ViewGroup) webView.  
            getParent()).removeView(webView);  
        webView.destroy();  
    }  
    super.onDestroy();  
}
```